反向传播 II

回顾：向量的导数

输入标量

输出标量

$x \in \mathbb{R}, y \in \mathbb{R}$

一般导数:

$$\frac{\partial y}{\partial x} \in \mathbb{R}$$

如果 x 变化了一个小量，y 会变化多少?

回顾：向量的导数

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输入**向量**
输出**标量**

$$x \in \mathbb{R}^N, y \in \mathbb{R}$$

导数是**梯度**

$$\frac{\partial y}{\partial x} \in \mathbb{R}^N \quad \left(\frac{\partial y}{\partial x}\right)_n = \frac{\partial y}{\partial x_n}$$

对于 x 的每个元素，如果它们变化了一个小量，那么 y 会变化多少？

回顾：向量的导数

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对于 x 的每个元素，如果它们变化了一个小量，那么 y 会变化多少?

输入**向量**
输出**向量**

$$x \in \mathbb{R}^N, y \in \mathbb{R}^M$$

导数是雅可比矩阵:

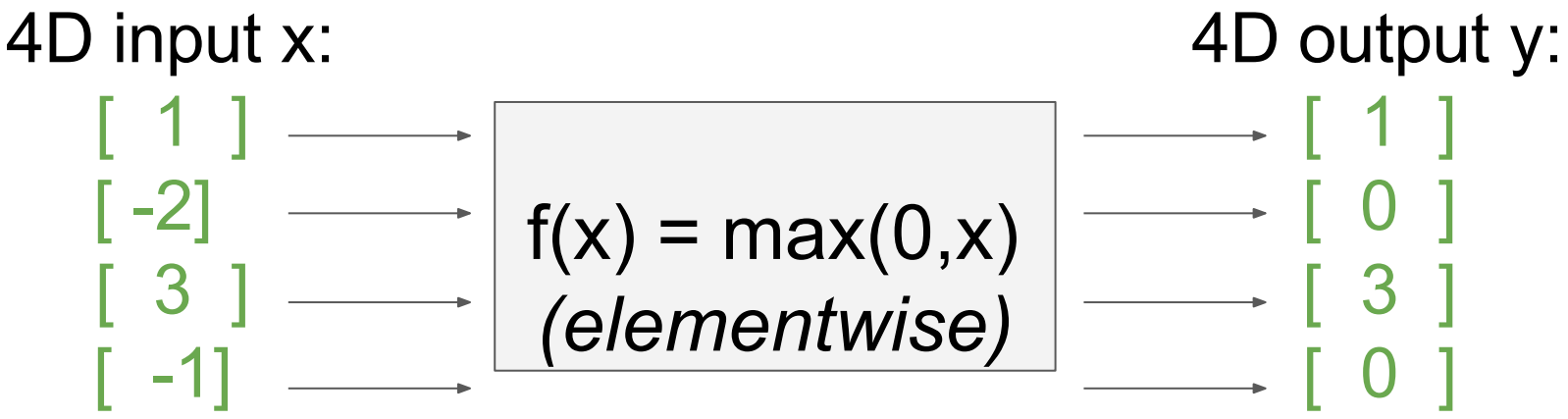
$$\frac{\partial y}{\partial x} \in \mathbb{R}^{N \times M} \quad \left(\frac{\partial y}{\partial x}\right)_{n,m} = \frac{\partial y_m}{\partial x_n}$$

对于 x 的每个元素，如果它变化了一个小量，那么 y 的每个元素会变化多少?

雅可比矩阵:梯度的推广

- 给定一个有m个输出和n个输入的函数
- $L(W) = [\mathrm{L}_1\ (w_1, w_2, \dots, w_n), \cdots, \mathrm{L}_m\ (w_1, w_2, \dots, w_n)]$
- 它的梯度（雅可比矩阵）是一个m x n的偏导数矩阵
- $\frac{\partial L}{\partial W} = \begin{bmatrix} \frac{\partial \mathrm{L}_1}{\partial w_1} & \cdots & \frac{\partial \mathrm{L}_1}{\partial w_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \mathrm{L}_m}{\partial w_1} & \cdots & \frac{\partial \mathrm{L}_m}{\partial w_n} \end{bmatrix}$

向量的后向传播



向量的后向传播

4D input x:

[1]
[-2]
[3]
[-1]

f(x) = max(0,x)
(elementwise)

4D output y:

[1]
[0]
[3]
[0]

4D dL/dy:

[4]
[-1]
[5]
[9]

上游梯度

向量的后向传播

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4D output y:

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[0]
[3]
[0]

雅可比矩阵 dy/dx

[1 0 0 0]
[0 0 0 0]
[0 0 1 0]
[0 0 0 0]

4D dL/dy:

[4]
[-1]
[5]
[9]

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向量的后向传播

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[dy/dx] [dL/dy]

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上游梯度

向量的后向传播

雅可比矩阵是**稀疏**的:非对角线项总是**零**!

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[0]
[3]
[0]

4D dL/dx:

[4]
[0]
[5]
[0]

[dy/dx] [dL/dy]

$$\left(\frac{\partial L}{\partial x}\right)_i = \begin{cases} \left(\frac{\partial L}{\partial y}\right)_i & \text{if } x_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

4D dL/dy:

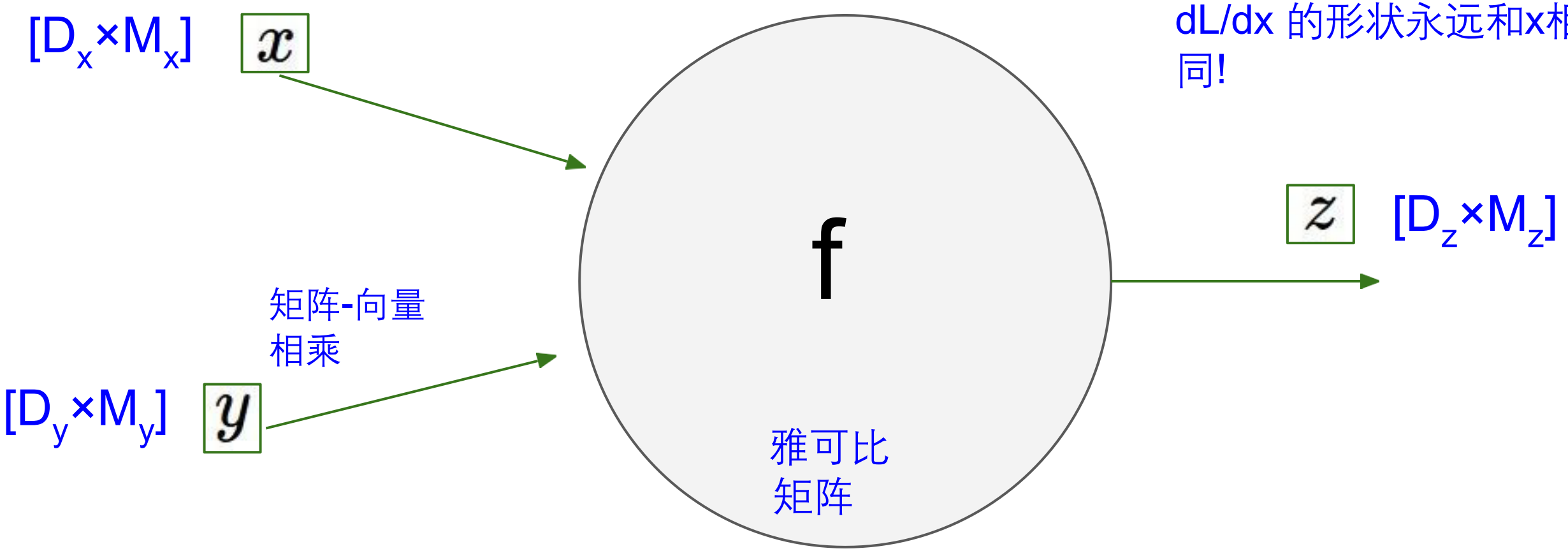
[4]
[-1]
[5]
[9]

上游梯度

矩阵 (或张量)的后向传播

Loss L 仍然是个标量!

dL/dx 的形状永远和x相同!



矩阵 (或张量)的后向传播

Loss L 仍然是个标量!

dL/dx 的形状永远和x相同!

$[D_x \times M_x]$ $\boxed{x}$

$[D_x \times M_x]$

$$\boxed{\frac{\partial L}{\partial x}} = \frac{\partial z}{\partial x}\frac{\partial L}{\partial x}$$

"下游梯度"

矩阵-向量相乘

"局部梯度"

$\boxed{\frac{\partial z}{\partial x}}$

$\boxed{\frac{\partial z}{\partial y}}$

雅可比矩阵

$\boxed{z}$ $[D_z \times M_z]$

$\boxed{\frac{\partial L}{\partial z}}$ $[D_z \times M_z]$

"上游梯度"

$[D_y \times M_y]$ $\boxed{y}$

$$\boxed{\frac{\partial L}{\partial y}} = \frac{\partial z}{\partial y}\frac{\partial L}{\partial z}$$

$[D_y \times M_y]$

对于y的每个元素，它对z的每个元素有多大的影响?

对于z的每个元素，它对L的影响有多大?

矩阵 (或张量)的后向传播

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dL/dx 的形状永远和x相同!

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矩阵-向量相乘

$[D_y \times M_y]$ $\boxed{y}$

$[D_y \times M_y]$

$$\boxed{\frac{\partial L}{\partial y}} = \frac{\partial z}{\partial y}\frac{\partial L}{\partial z}$$

"局部梯度"

$\boxed{\frac{\partial z}{\partial x}}$ $[(D_x \times M_x) \times (D_z \times M_z)]$

$\boxed{\frac{\partial z}{\partial y}}$ $[(D_y \times M_y) \times (D_z \times M_z)]$

雅可比矩阵

$\boxed{z}$ $[D_z \times M_z]$

$\boxed{\frac{\partial L}{\partial z}}$ $[D_z \times M_z]$

"上游梯度"

对于y的每个元素，它对z的每个元素有多大的影响?

对于z的每个元素，它对L的影响有多大?

矩阵的后向传播

2 个样本，每个样本 3 维

x: [N×D]

$$x = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix}$$

w: [D×M]

$$w = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & w_{23} & w_{24} \\ w_{31} & w_{32} & w_{33} & w_{34} \end{bmatrix}$$

3 输入 → 4 输出

矩阵乘法

$$y_{n,m} = \sum_d x_{n,d} w_{d,m}$$

y: [N×M]

$$y = \begin{bmatrix} y_{11} & y_{12} & y_{13} & y_{14} \\ y_{21} & y_{22} & y_{23} & y_{24} \end{bmatrix}$$

第一个样本： [2 1 -3]

$$y_{11} = x_{11}w_{11} + x_{12}w_{21} + x_{13}w_{31}$$
$$y_{12} = x_{11}w_{12} + x_{12}w_{22} + x_{13}w_{32}$$
$$y_{13} = x_{11}w_{13} + x_{12}w_{23} + x_{13}w_{33}$$
$$y_{14} = x_{11}w_{14} + x_{12}w_{24} + x_{13}w_{34}$$

第二个样本： [-3 4 2]

$$y_{21} = x_{21}w_{11} + x_{22}w_{21} + x_{23}w_{31}$$
$$y_{22} = x_{21}w_{12} + x_{22}w_{22} + x_{23}w_{32}$$
$$y_{23} = x_{21}w_{13} + x_{22}w_{23} + x_{23}w_{33}$$
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构建雅可比矩阵 $\frac{\partial y}{\partial x}$

- y 有 8 个元素（2 行 × 4 列）→ 行数：8
- x 有 6 个元素（2 行 × 3 列）→ 列数：6
- 所以雅可比矩阵是： $\frac{\partial y}{\partial x} \in \mathbb{R}^{8\times 6}$

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$$w = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & w_{23} & w_{24} \\ w_{31} & w_{32} & w_{33} & w_{34} \end{bmatrix}$$

$$\frac{\partial y_1}{\partial x_1} = w^T \in \mathbb{R}^{4\times 3}$$

	x_{11}	x_{12}	x_{13}	x_{21}	x_{22}	x_{23}
y_{11}	w_{11}	w_{21}	w_{31}	0	0	0
y_{12}	w_{12}	w_{22}	w_{32}	0	0	0
y_{13}	w_{13}	w_{23}	w_{33}	0	0	0
y_{14}	w_{14}	w_{24}	w_{34}	0	0	0
y_{21}	0	0	0	w_{11}	w_{21}	w_{31}
y_{22}	0	0	0	w_{12}	w_{22}	w_{32}
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y_{14}	w_{14}	w_{24}	w_{34}	0	0	0
y_{21}	0	0	0	w_{11}	w_{21}	w_{31}
y_{22}	0	0	0	w_{12}	w_{22}	w_{32}
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$$w = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & w_{23} & w_{24} \\ w_{31} & w_{32} & w_{33} & w_{34} \end{bmatrix}$$

$$\frac{\partial y_2}{\partial x_2} = w^T \in \mathbb{R}^{4\times 3}$$

构建雅可比矩阵 $\frac{\partial y}{\partial x}$

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$\frac{\partial y}{\partial x} =$

	x_{11}	x_{12}	x_{13}	x_{21}	x_{22}	x_{23}
y_{11}	w_{11}	w_{21}	w_{31}	0	0	0
y_{12}	w_{12}	w_{22}	w_{32}	0	0	0
y_{13}	w_{13}	w_{23}	w_{33}	0	0	0
y_{14}	w_{14}	w_{24}	w_{34}	0	0	0
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y_{23}	0	0	0	w_{13}	w_{23}	w_{33}
y_{24}	0	0	0	w_{14}	w_{24}	w_{34}

$$= \begin{bmatrix} w^T & 0 \\ 0 & w^T \end{bmatrix} \in \mathbb{R}^{8\times 6}$$

矩阵的后向传播

2 个样本，每个样本 3 维

x: [N×D]

[2 1 -3]
[-3 4 2]

w: [D×M]

[3 2 1 -1]
[2 1 3 2]
[3 2 1 -2]

3 输入 → 4 输出

矩阵乘法

$$y_{n,m} = \sum_d x_{n,d} w_{d,m}$$

y: [N×M]

[-1 -1 2 6]
[5 2 11 7]

dL/dy: [N×M]

[2 3 -3 9]
[-8 1 4 6]

我们不会真的构造这个大块矩阵!

而是使用:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} \cdot w^T$$

这样做等价于:

- 对每个样本 i，计算:

$$\frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y_i} \cdot w^T$$

矩阵的后向传播

x: [N×D]

[2 **1** -3]
[-3 4 2]

w: [D×M]

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矩阵乘法

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[**-1** **-1** **2** **6**]
[5 2 11 7]

dL/dy: [N×M]

[2 3 -3 9]
[-8 1 4 6]

- 雅可比矩阵:
  - dy/dx: [(N×D)×(N×M)]
  - dy/dw: [(D×M)×(N×M)]

矩阵的后向传播

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[2 **1** -3]
[-3 4 2]

w: [D×M]

[3 2 1 -1]
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Q: y的哪些部分会受到x的一个元素的影响?

矩阵的后向传播

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Q: y的哪些部分会受到x的一个元素的影响?

A: $x_{n,d}$ 影响整个列 $y_{n,\cdot}$

$$\frac{\partial L}{\partial x_{n,d}} = \sum_m \frac{\partial L}{\partial y_{n,m}} \frac{\partial y_{n,m}}{\partial x_{n,d}}$$

矩阵的后向传播

x: [N×D]

[2 1 -3]
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w: [D×M]

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A: $x_{n,d}$ 影响整个列 $y_{n,\cdot}$

Q: $x_{n,d}$ 会影响 $y_{n,m}$ 多少？

$$\frac{\partial L}{\partial x_{n,d}} = \sum_m \frac{\partial L}{\partial y_{n,m}} \frac{\partial y_{n,m}}{\partial x_{n,d}}$$

矩阵的后向传播

x: [N×D]

[2 1 -3]
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w: [D×M]

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dL/dy: [N×M]

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Q: y的哪些部分会受到x的一个元素的影响？

A: $x_{n,d}$ 影响整个列 $y_{n,\cdot}$

Q: $x_{n,d}$ 会影响 $y_{n,m}$ 多少？

A: $w_{d,m}$

[N×D][N×M] [M×D]

$$\frac{\partial L}{\partial x} = \left(\frac{\partial L}{\partial y}\right) w^T$$

$$\frac{\partial L}{\partial x_{n,d}} = \sum_m \frac{\partial L}{\partial y_{n,m}} \frac{\partial y_{n,m}}{\partial x_{n,d}} = \sum_m \frac{\partial L}{\partial y_{n,m}} w_{d,m}$$

矩阵的后向传播

x: [N×D]

[2 1 -3]
[-3 4 2]

w: [D×M]

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矩阵乘法

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dL/dy: [N×M]

[2 3 -3 9]
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[N×D][N×M] [M×D]

$$\frac{\partial L}{\partial x} = \left(\frac{\partial L}{\partial y}\right) w^T$$

[D×M] [D×N] [N×M]

$$\frac{\partial L}{\partial w} = x^T \left(\frac{\partial L}{\partial y}\right)$$

这些公式很容易记住:它们是使形状匹配的唯一方法。

矩阵的后向传播

x: [N×D]

[2 1 -3]
[-3 4 2]

w: [D×M]

[3 2 1 -1]
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矩阵乘法

$$y_{n,m} = \sum_d x_{n,d} w_{d,m}$$

y: [N×M]

[-1 -1 2 6]
[5 2 11 7]

dL/dy: [N×M]

[2 3 -3 9]
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$$w^T = \begin{bmatrix} 3 & 2 & 3 \\ 2 & 1 & 2 \\ 1 & 3 & 1 \\ -1 & 2 & -2 \end{bmatrix}$$

[N×D][N×M] [M×D]

$$\boxed{\frac{\partial L}{\partial x} = \left(\frac{\partial L}{\partial y}\right) w^T}$$

$$\frac{\partial L}{\partial x} = \begin{bmatrix} 2 & 3 & -3 & 9 \\ -8 & 1 & 4 & 6 \end{bmatrix} \cdot \begin{bmatrix} 3 & 2 & 3 \\ 2 & 1 & 2 \\ 1 & 3 & 1 \\ -1 & 2 & -2 \end{bmatrix}$$

矩阵的后向传播

x: [N×D]

[2 1 -3]
[-3 4 2]

w: [D×M]

[3 2 1 -1]
[2 1 3 2]
[3 2 1 -2]

矩阵乘法

$$y_{n,m} = \sum_d x_{n,d} w_{d,m}$$

y: [N×M]

[-1 -1 2 6]
[5 2 11 7]

dL/dy: [N×M]

[2 3 -3 9]
[-8 1 4 6]

$$x^T = \begin{bmatrix} 2 & -3 \\ 1 & 4 \\ -3 & 2 \end{bmatrix}$$

[D×M] [D×N] [N×M]

$$\frac{\partial L}{\partial w} = x^T \left(\frac{\partial L}{\partial y} \right)$$

$$\frac{\partial L}{\partial w} = \begin{bmatrix} 2 & -3 \\ 1 & 4 \\ -3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 3 & -3 & 9 \\ -8 & 1 & 4 & 6 \end{bmatrix}$$

总结

- **(全连接)神经网络**是一系列线性函数和非线性激活函数的堆叠；它们比线性分类器具有更强的表征能力。
- **反向传播** = 沿着计算图的递归应用链式法则来计算所有输入/参数/中间值的梯度
- **前向传播：**计算每个操作的结果，并将梯度计算所需的任何中间值保存在内存中
- **反向传播：**应用链式法则来计算损失函数相对于输入的梯度

ConvNetJS demo: Classify toy 2D data (stanford.edu)

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$

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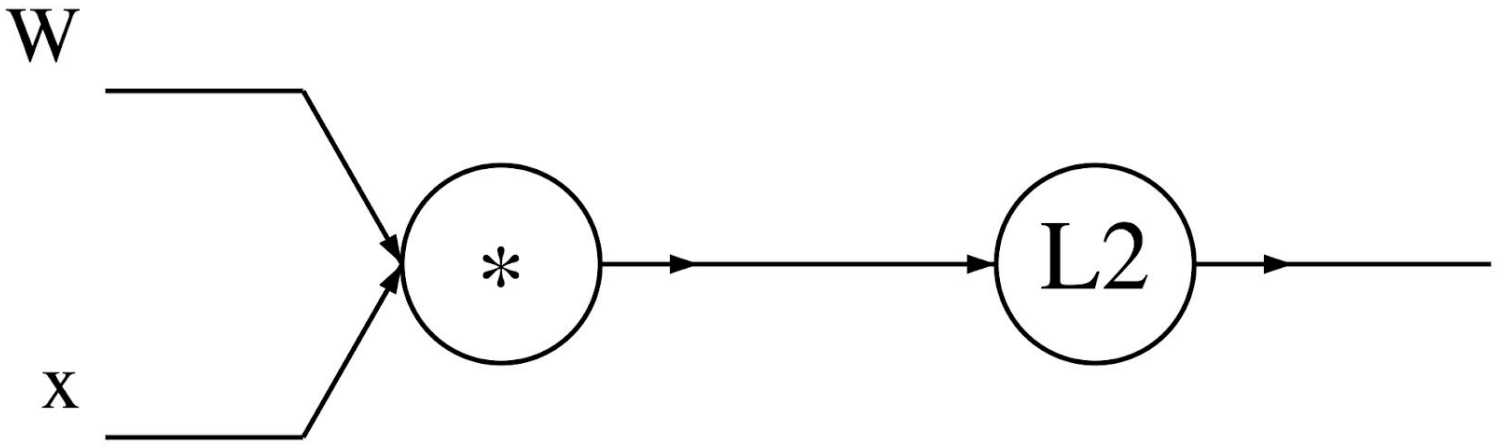
$$\in \mathbb{R}^n \quad \in \mathbb{R}^{n \times n}$$

A vectorized example: $f(x, W) = ||W \cdot x||^2 = \sum_{i=1}^{n} (W \cdot x)_i^2$



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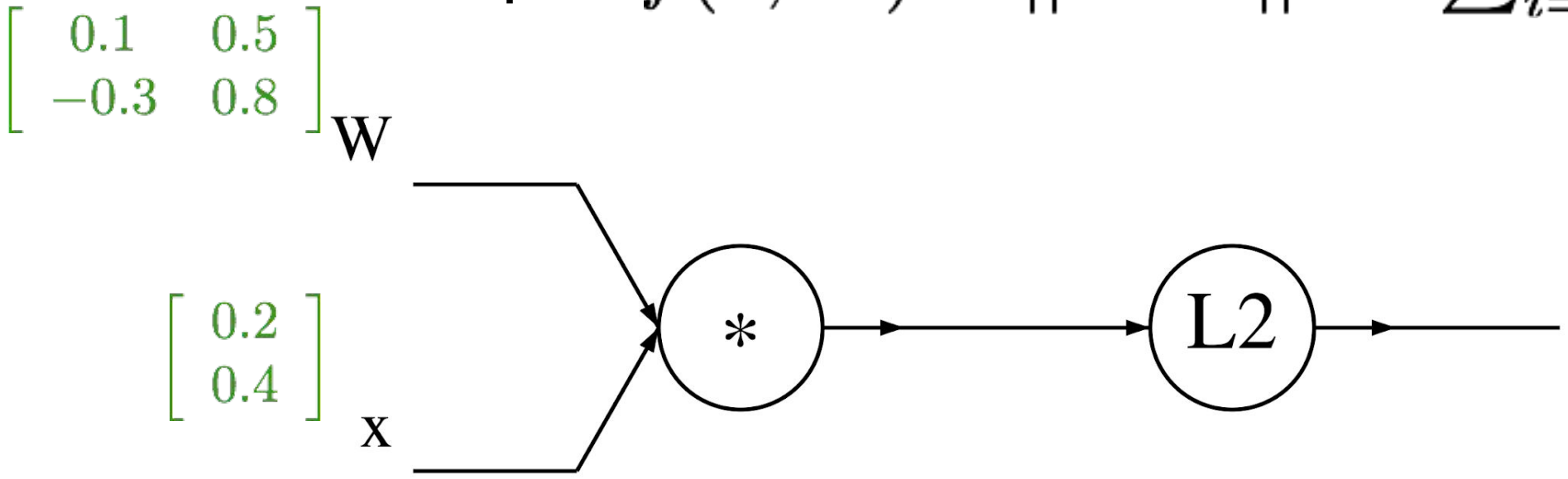

$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = ||q||^2 = q_1^2 + \cdots + q_n^2$$

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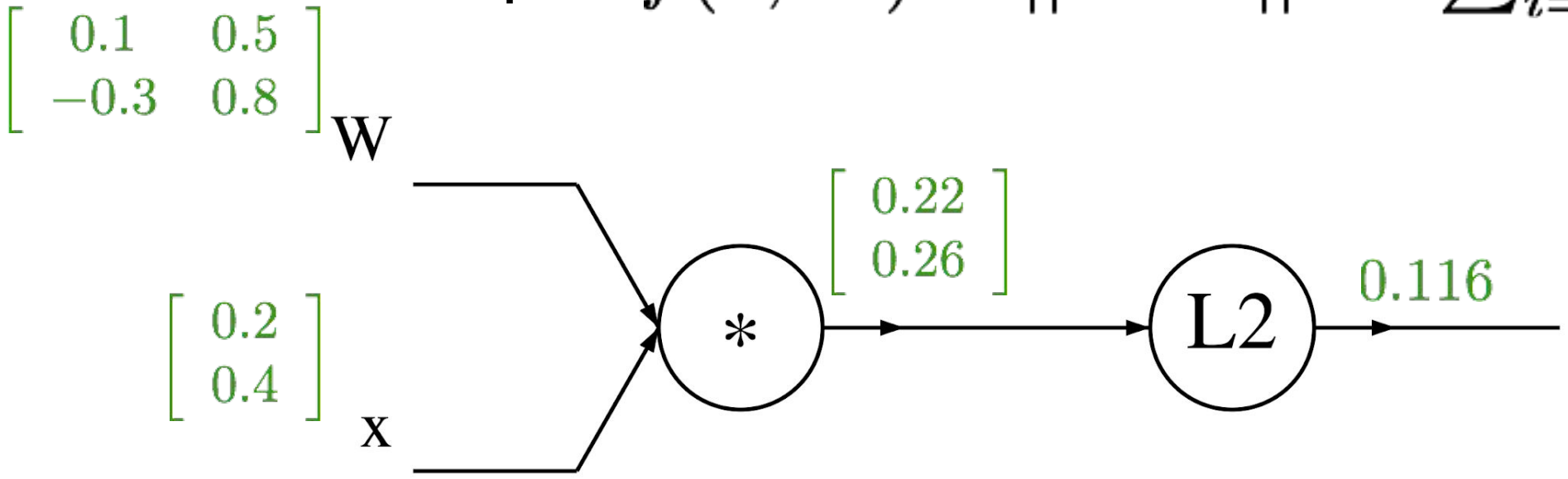

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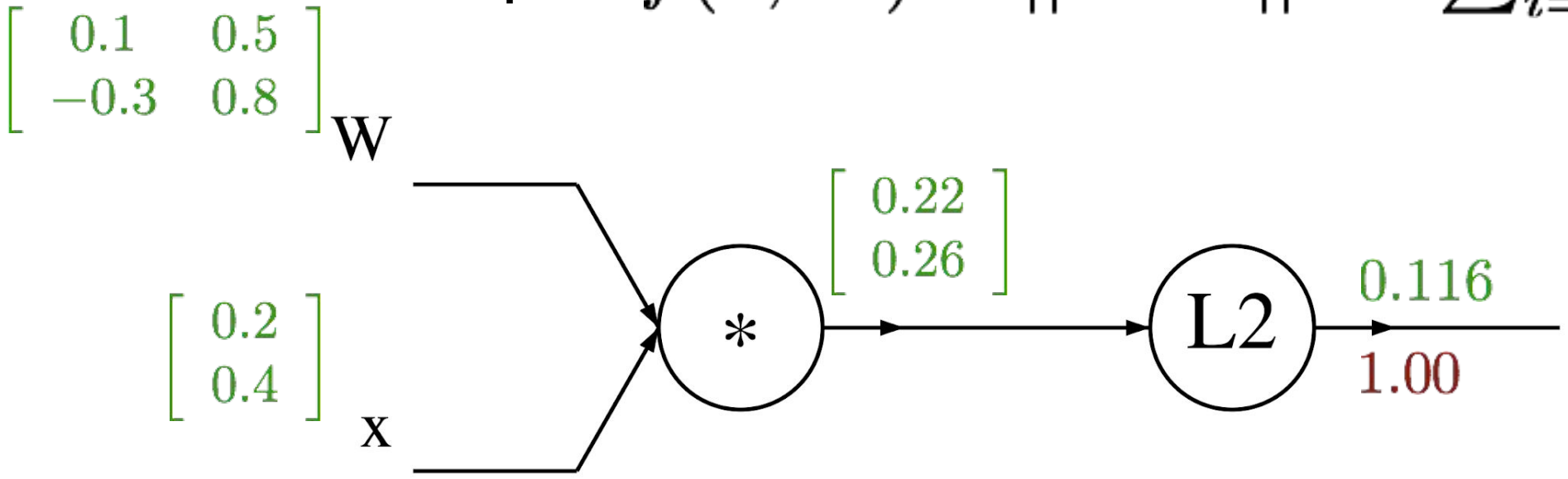

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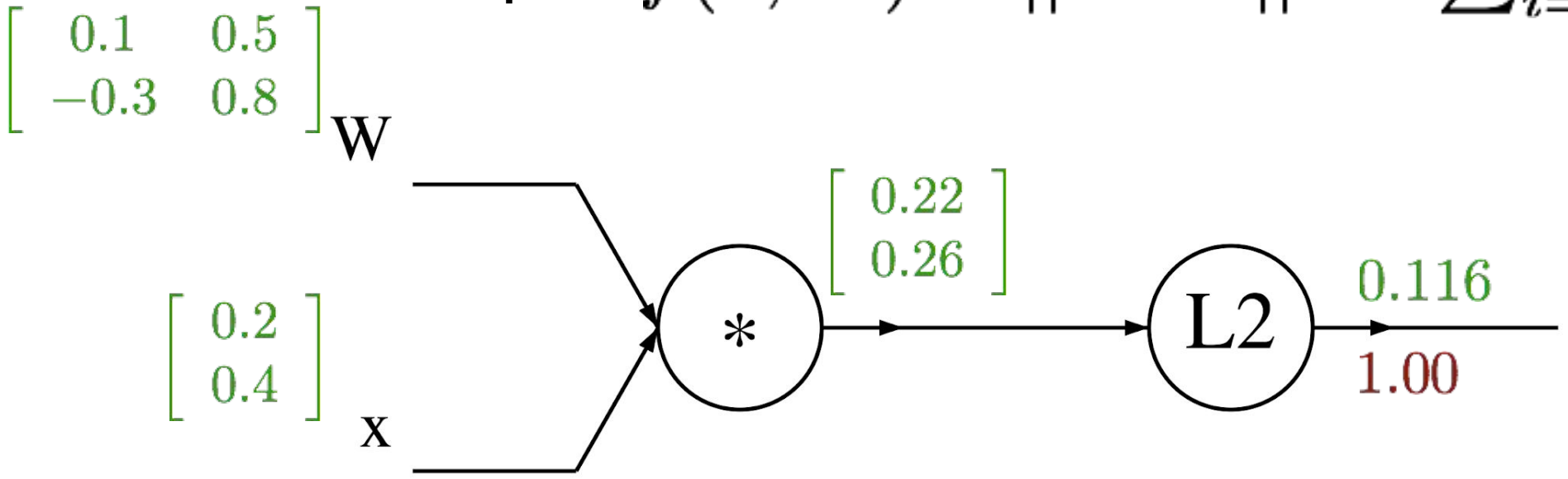

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$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\boxed{\nabla_q f = 2q}$$

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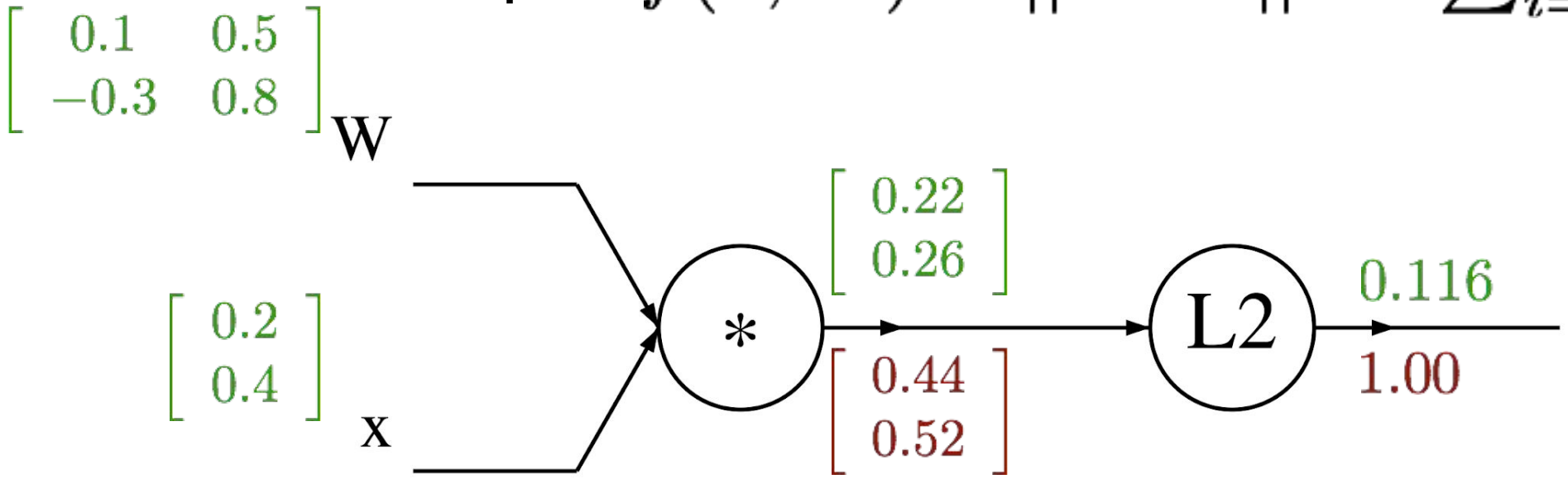

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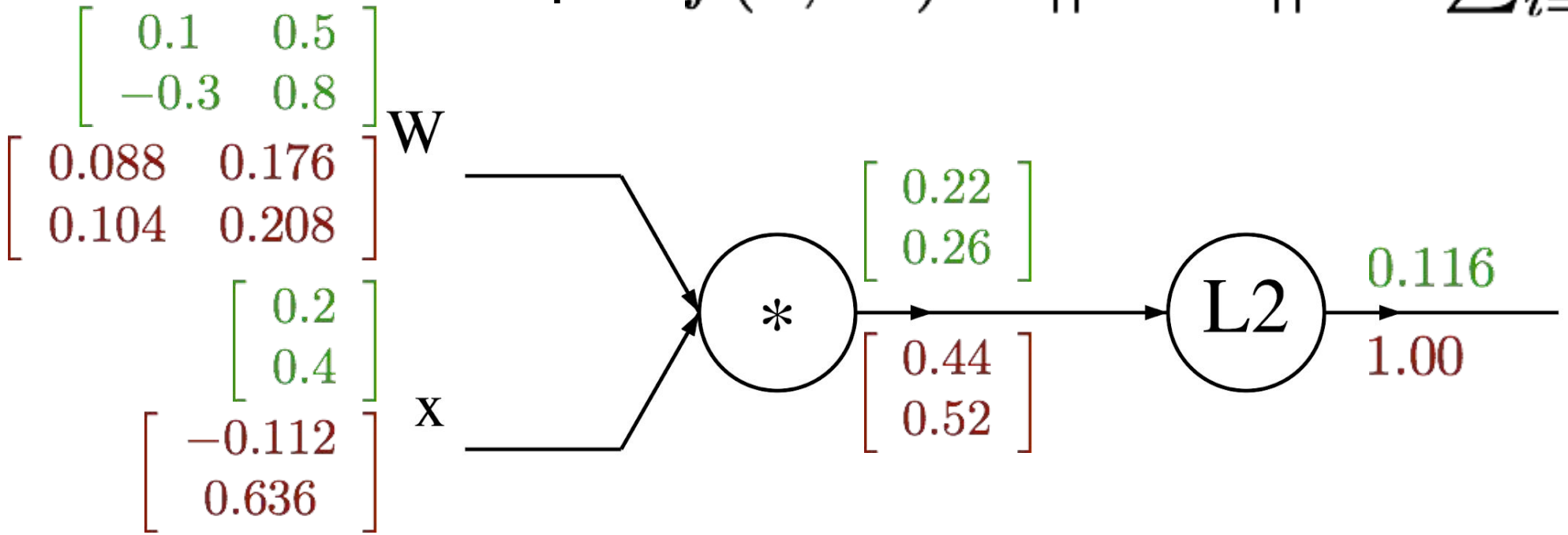

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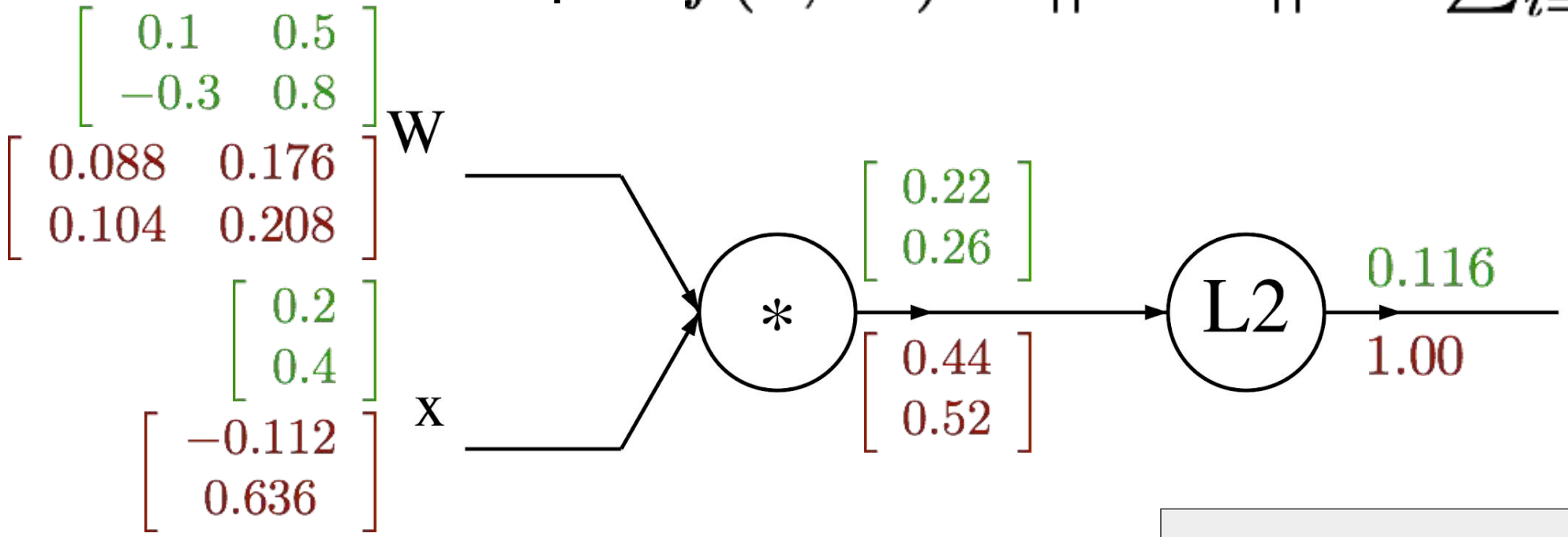

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$$f(q) = ||q||^2 = q_1^2 + \cdots + q_n^2$$

$$\boxed{\nabla_x f = 2W^T \cdot q}$$